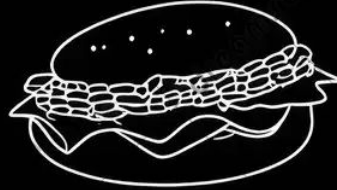
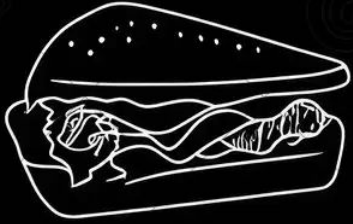
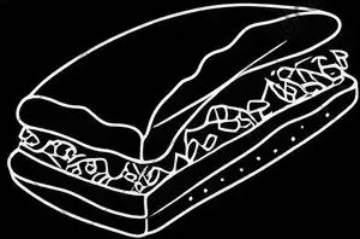


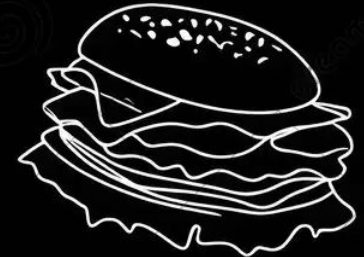
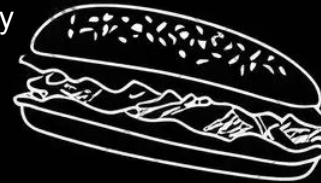
Classifying Foods by Image



Modelling with Advanced Machine Learning Methods



Group 9
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Andre Ghandt
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Goals

- Build visual learning models for food classification
- Compare the performance of different training strategies
- Determine how models perform on modified test sets
- Determine the most accurate and robust model
- Analyze which regions the models focus on

Food-101 Dataset

- 750 training images for each of 101 food categories
- 6 test sets, 250 test images per category for each
 - 1 clean test set
 - 5 test set variants

sushi

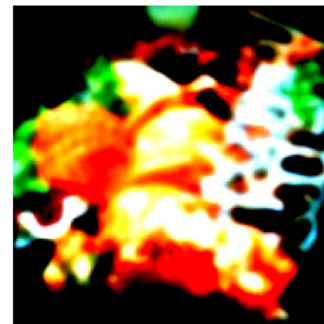
Clean



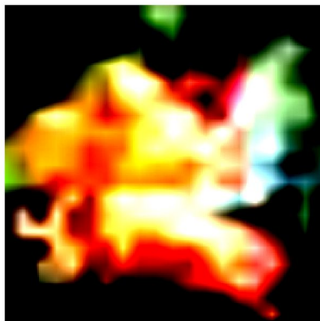
Blur Little



Blur Medium



Downsampled



Masked



Noise Rotation



Models

2 backbones with ImageNet-1K pretrained weights:

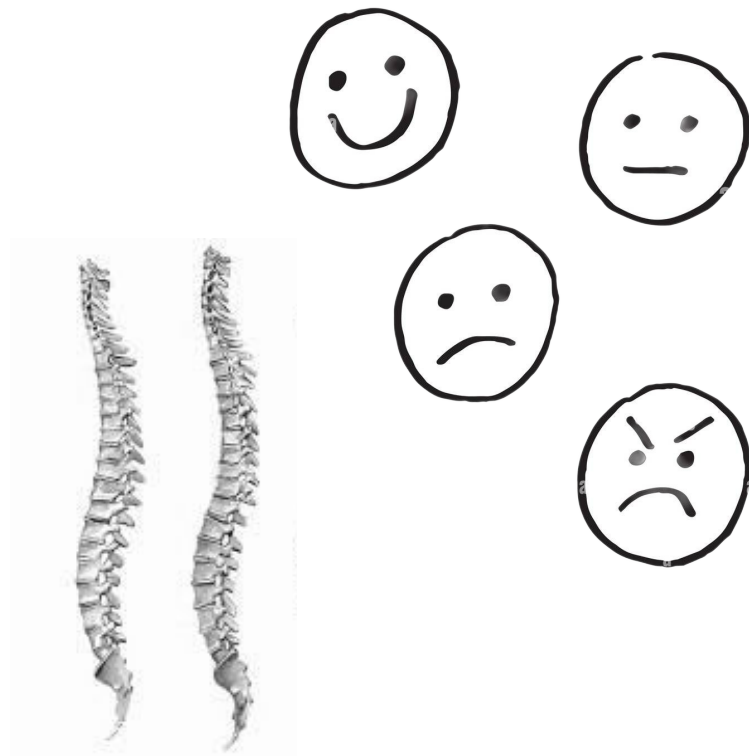
- ResNet18
- EfficientNetV2-S

4 heads using different training strategies:

- Linear Probing (baseline)
- LoRA
- Task-Specific Adapters
- BatchNorm Tuning

2 backbones $\times$ 4 heads = 8 models

- 5 epochs each



Evaluations

8 models $\times$ 6 test sets = 48 evaluations

Method	Backbone	Accuracy	Avg Robustness
Linear Probing	ResNet18	0.492	0.704
Linear Probing	EfficientNetV2-S	0.453	0.681
LoRA	ResNet18	0.712	0.755
LoRA	EfficientNetV2-S	0.822	0.809
Task-Specific Adapters	ResNet18	0.574	0.716
Task-Specific Adapters	EfficientNetV2-S	0.661	0.749
BatchNorm Tuning	ResNet18	0.659	0.731
BatchNorm Tuning	EfficientNetV2-S	0.812	0.805

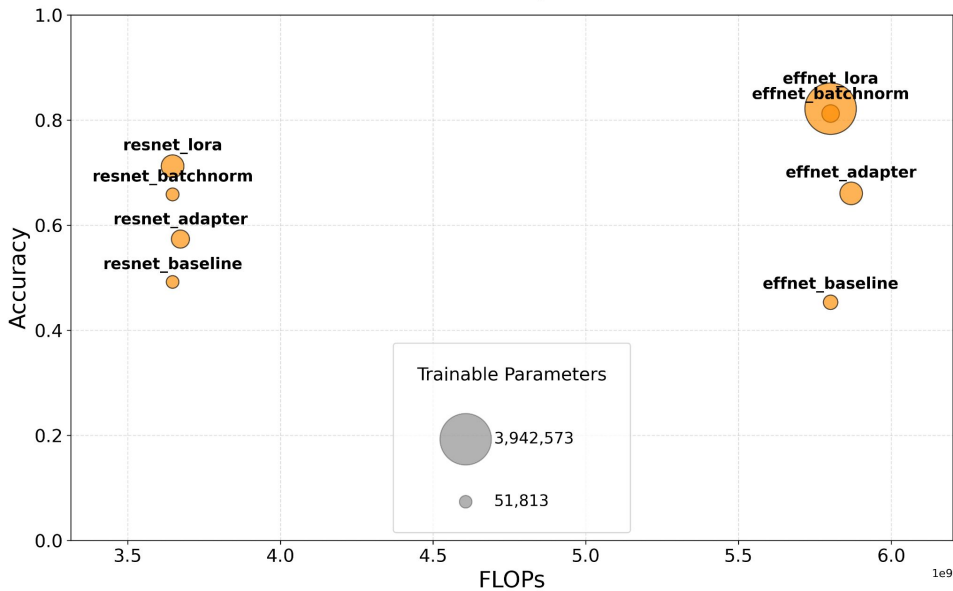
Comparing Test Variants

Results for best model (LoRA with EfficientNetV2-S)

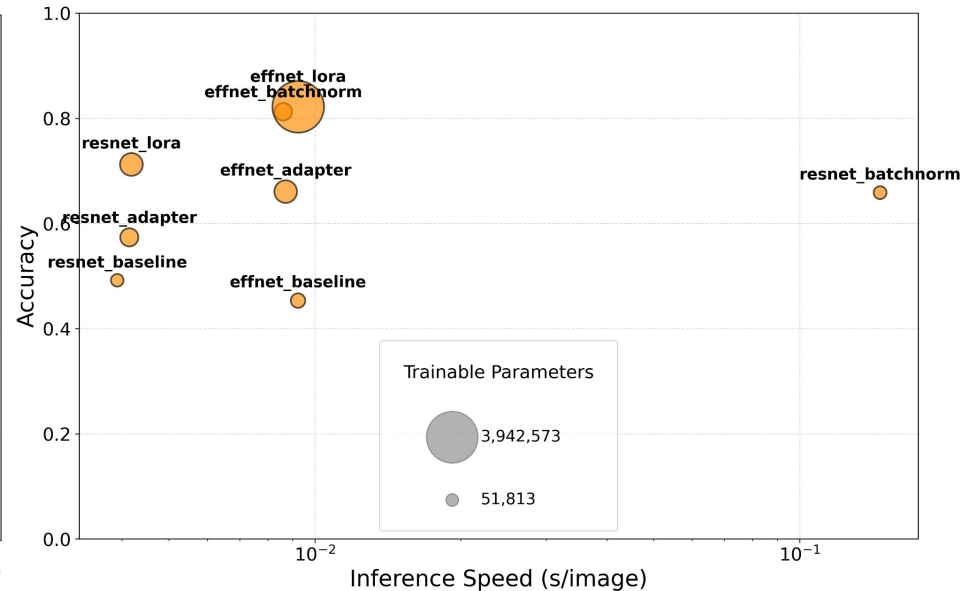
Test Variant	Accuracy	Robustness
Clean	0.822	1.000
Masked	0.815	0.991
Noise Rotation	0.786	0.956
Blur Little	0.758	0.923
Blur Medium	0.619	0.753
Downsampled	0.346	0.422

Complexity and Speed

Model Accuracy vs. FLOPs



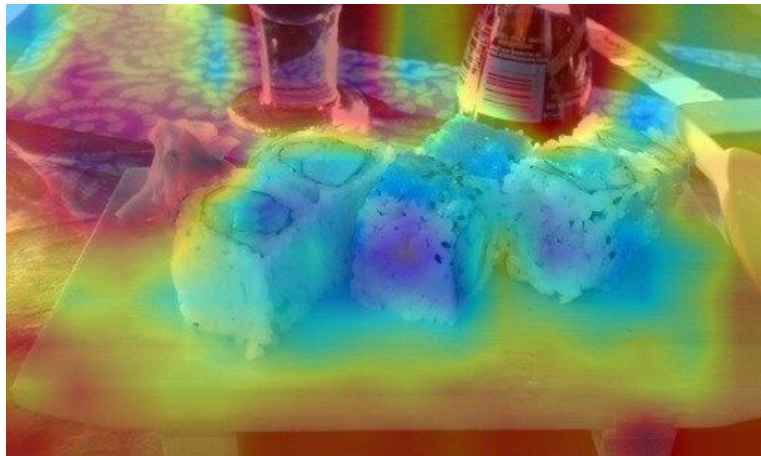
Model Accuracy vs. Inference Speed



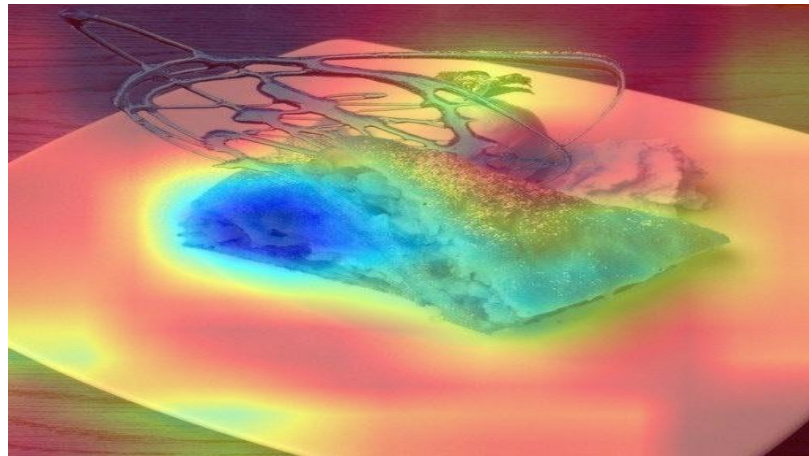
Model Focus

- Application of grad-CAM
- Applied to a sample of 5 images from each test set for each model
- Interpret heatmaps to discover patterns amongst different models for different test variants

Classification often highly influenced by background



Effnet Adapter



Resnet Baseline (Linear Probing)

This effect appears to be more pronounced in models with fewer active layers during the downstream fine-tuning of a pretrained model (like linear probing).

Downsampled Variant has lowest accuracy amongst all models



Resnet Batchnorm



Effnet Batchnorm

We can observe from the heatmaps that the models fail to identify spatial structure in the downsampled images. This suggests that image size and fewer pixels makes it more difficult to identify spatial information. More pixels allow CNN's to more accurately identify details.

Future work?

- We are underfitting, so training with more than 5 epochs would help
- Investigate inference speed outlier
- Hyperparameter tuning

